

[DC] Dynamic facial expressions on virtual humans for virtual reality (VR) mental health therapy

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Figure 1: Virtual human with the facial expressions from embarrassed to happy by blendshapes interpolation

ABSTRACT

This PhD research aims to implement dynamic facial expressions on virtual humans and explore their potential to enhance the efficacy of virtual reality (VR) mental health therapy. A systematic review of virtual humans in mental health VR indicated that only around 10% of applications used dynamic facial expressions. The potential of virtual humans' emotion richness is understudied and it is unclear how the facial expressions affect the individuals in VR. Therefore, we will focus on understanding people's behavioural, physiological, and psychological reactions toward facial-animated humans in VR experimental studies. The first study examines whether particular non-verbal behaviours can enhance people's therapy engagement, by applying warm facial expressions and head nods on a virtual coach. Future experiments will further look at individuals' interpretations of facial expressions on virtual crowds and the virtual infant. This research will explore how best to use facial expressions to facilitate VR therapy through the practice of psychiatric research, VR programming and 3D animation.

Index Terms: Human-centred computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Creating expressive virtual humans for social interaction is vital for the development of VR mental health therapy. As a key component of the virtual human, the use of facial animation can increase the perceived realism and influence user's attitude towards the social interaction in virtual environment ([4, 6, 7]). However, due to the complexity of facial expressions modelling and limited understanding of how emotional facial expressions are processed in VR, most mental health applications remain utilizing the neutral-face human characters rather than introducing emotional facial expressions. It is unclear how best to apply facial expressions for people with different mental health states. This understudied element sets the challenge for the VR content creators and psychologists to enhance the quality and efficacy of VR therapies.

The proposed research, therefore, focuses on using facial expressions on virtual human to affect people's responses and their willingness to face their fears in VR. Dynamic facial expressions will be implemented to understand people's behavioural, physiological, and psychological reactions toward facial-animated characters. We

will also look at the individual difference by taking participants' own mental health states (e.g., mistrust tendencies) into consideration.

1. Understand the uses and potentials of virtual humans and their facial expressions in immersive VR to advance mental health research.
2. Create and implement context-related facial animations on virtual humans in VR.
3. Investigate how virtual humans' dynamic facial expressions affect participants' behaviours and attitudes in VR.
4. Examine the potential moderated effect of people's mental health states on their responses towards facial-animated virtual humans in VR.

2 CURRENT STUDIES

2.1 Systematic review of virtual humans in mental health VR

Virtual human creation is a complex task that involves multiple aspects in psychology and computer science, yet little systematic work has focused on virtual humans and their traits in mental health VR. To establish a fundamental overview of virtual humans, a systematic review was conducted to highlight their current use and future opportunities to advance mental health research.

55 out of 1181 potential eligible studies (using search terms relating to virtual reality, virtual human and mental health disorders) are selected for review. Virtual humans are used extensively for mental health research, especially for social anxiety ($n = 18$), eating disorders ($n = 15$) and psychosis and schizophrenia ($n = 10$). They function as active social interaction partners (24/54), virtual coaches (3/54), virtual crowds (13/54) and virtual body reference (15/54). They are mainly controlled by computer (agent) rather than by person in real-time (avatar). Virtual human's visual appearance (body size, gender and ethnicity) are the major traits for experimental manipulation ($n = 20$). Their behaviours ($n = 18$) such as responsive level and gaze patterns are also explored. The impact of virtual humans' personality ($n=1$) and facial expressions ($n = 2$) are lack of study.

Traits of virtual humans have different levels of impact, and the potential effect of virtual human's emotional expression is still to be discovered. We consider virtual humans as a rapidly developing area and a key solution to understanding complex human behaviours in mental health VR. Researchers need to better address the contributions of virtual humans in mental health study and provide a comprehensive presentation about virtual characters to communicate research outcomes.

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2.2 Experimental study: Facial expressions on a VR coach

This ongoing study aims to examine whether particular non-verbal behaviours of a virtual coach can enhance people's therapy engagement in VR.

Virtual coaches are starting to be used in automated virtual reality therapy for mental health difficulties. They provide instruction, education, and modelling behaviours and therefore, successful engagement with virtual coaches can potentially affect therapy outcomes [3, 5]. Although evidence suggests that positive non-verbal communication such as smiles and body movement can increase the perceived friendliness and trust of virtual humans, little is known regarding the virtual coach in VR therapy. The study will specifically look at the impact of a virtual coach's warm facial expression and affirmative head nod on people's therapeutic alliance and perceived credibility and expectancy of the VR treatment. Additionally, we are also interested to explore the association between participants' levels of mistrust and their alliance with the VR coach.

2.2.1 Questions and hypothesis

- Primary question: Does the addition of positive facial expression and/or head movement enhance (a) the therapeutic alliance with a VR coach and (b) the perceived credibility and expectancy of the VR treatments?
- Secondary question: Is there any moderated effect of participant's mistrust tendency on their therapeutic alliance with the virtual coach?
- H_p : The addition of warm facial expression and/or head nod can lead to the (a) higher therapeutic alliance with a VR coach and (b) higher perceived credibility and expectancy of the VR treatment. The combined use of warm facial expression and head nod will yield the strongest positive effect.
- H_s : Participants' mistrust tendencies have a negative effect on their therapeutic alliance with the virtual coach in VR.

2.2.2 Methods



Figure 2: VR coach in the consultation room

An introductory session of the VR fear of heights consultation (figure 2) and virtual height experience are developed for empirical testing. We are looking at the impact of 2 factors about virtual coach: **warmness facial expression** (with/without warmness facial expression - neutral face) and **head nod** (with/without head nod).

Participants will be randomised to one of four conditions, to interact with a virtual coach has (1) neutral face (2) neutral face and head nod (3) warmness facial expression (4) warmness facial expression and head nod. In all these conditions the virtual coach will have the basic facial movement including eye blink and lip sync.

After the VR consultation, participants will also be asked to try out a virtual height experience in VR.

Eligible participants who have a fear of heights will be invited for the VR experiment.

2.2.3 Measures

- Therapeutic alliance: Participant's therapeutic alliance towards virtual coach will be measured by The Virtual Therapist Alliance Scale (VTAS) [5]
- Willingness to continue: Participants' willingness to continue the VR fear of heights treatment will be measured by the Credibility/expectancy questionnaire [1]
- Mistrust tendencies: Participants' mistrust level will be measured by the The Revised Green et al., Paranoid Thoughts Scale (R-GPTS) [2]
- Physiological response: Participants' physiological data (heart rate and galvanic skin response) will be collected for supplementary insights of arousal level.

3 FUTURE RESEARCH

With the systematic review and the ongoing virtual coach study, we aim to understand the current state of virtual humans, and the impacts of their facial expressions in mental health VR. Future studies will continue to look at the individual difference in interpreting the dynamic facial expressions in VR. We are interested to examine:

- Whether the dynamic facial expressions on virtual crowds affect people's interpretation of VR scenarios.
- The feasibility to create dynamic facial expressions on a virtual baby to prepare people with mental health difficulties for parenting activities.

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